

Homework 4

Lattice planes, XRD, and free electrons

1. (24pts) Here we will explore different families of lattice planes in Graphene. Below I have a picture of the crystal structure and the reciprocal space lattice for your convenience. Please find the first three families of inequivalent lattice planes (first being the largest spacing). More specifically:
 - (a) Please extract the spacing of each family of planes from the appropriate reciprocal lattice vector.
 - (b) Sketch the planes in real space.
 - (c) Show the the spacing obtained geometrically from the real space lattice does indeed match what you deduce from the respective reciprocal lattice vector.

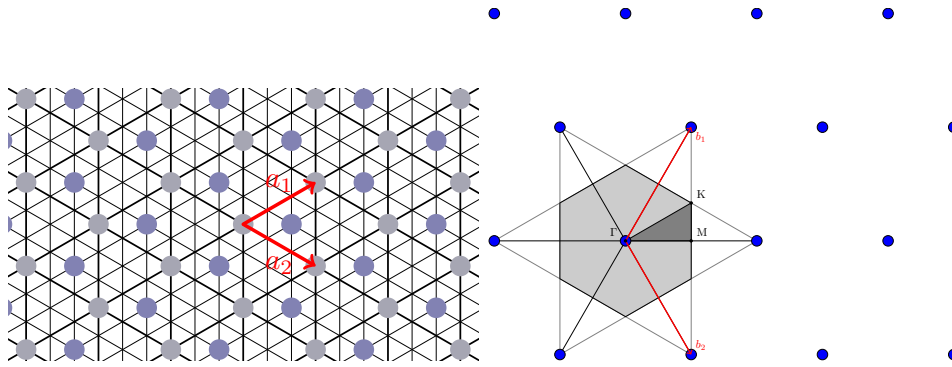
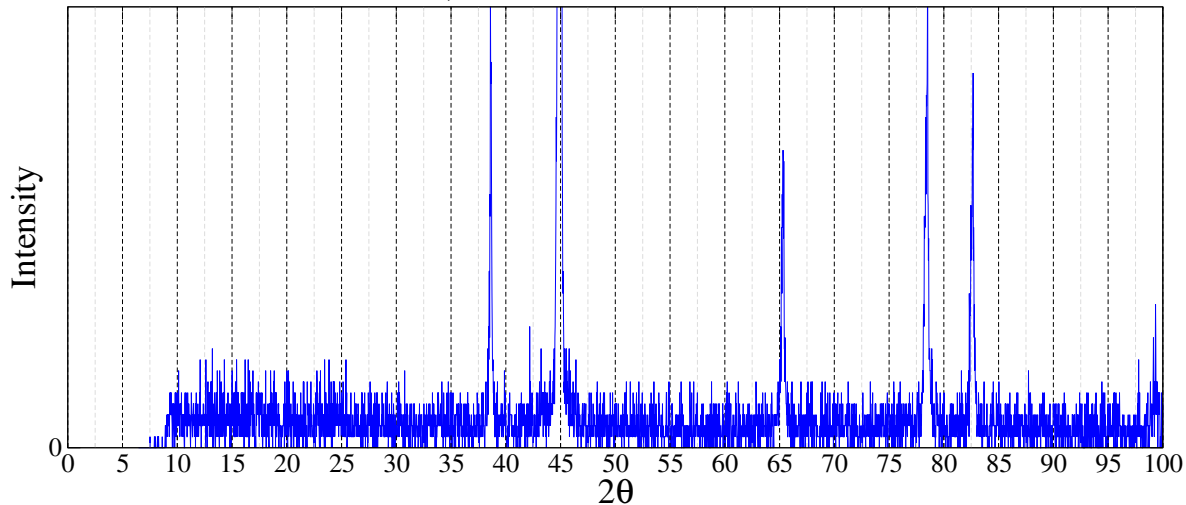


Figure 1: (Left panel) Crystal structure of graphene. The lattice parameters are given as $\vec{a}_1 = a\frac{\sqrt{3}}{2}\vec{i} + a\frac{1}{2}\vec{j}$ and $\vec{a}_2 = a\frac{\sqrt{3}}{2}\vec{i} - a\frac{1}{2}\vec{j}$. (Right panel) The reciprocal lattice; the lattice parameters can be derived from the real space vectors; and they were given in class.

2. (38pts) Below we have a powder x-ray diffraction measurement for a FCC material containing a single type of atom; the wavelength of the x-rays is $\lambda = 1.54\text{\AA}$. Assuming a cubic lattice with a basis, index the first five peaks shown in the plot. Also, extract the lattice parameter and guess which material this is (ie. look up table with FCC lattice parameters and compare.)



3. (38pts) This question explore free electrons. Assume an “empty lattice” which is two dimensional square lattice with lattice constant a . Plot the electronic band structure along a path in the first Brillouin zone according to the following lattice coordinates in reciprocal space: $(0, 0) \rightarrow (\frac{1}{2}, 0) \rightarrow (\frac{1}{2}, \frac{1}{2}) \rightarrow (0, 0)$. Be sure to include at least the four lowest bands at each k -point, and label each branch according to the reciprocal lattice vector.